

证明: (a) 推导正则对易关系

已知 $[\hat{x}, \hat{p}_x] = i\hbar$, 且不同分量间对易, 如 $[\hat{x}, \hat{p}_y] = 0$, 可推广到三维:

- 同一分量的位置和动量: $[\hat{r}_i, \hat{p}_j] = i\hbar \delta_{ij}$ ($i, j = x, y, z$, δ_{ij} 是克罗内克函数, $i=j$ 时为 1, 否则为 0), 如 $[\hat{x}, \hat{p}_x] = i\hbar$, $[\hat{y}, \hat{p}_y] = i\hbar$ 等
- 不同分量的位置和动量: 位置分量间对易 $[\hat{r}_i, \hat{r}_j] = 0$, (如 $[\hat{x}, \hat{y}] = 0$); 动量分量间对易 $[\hat{p}_i, \hat{p}_j] = 0$, (如 $[\hat{p}_x, \hat{p}_y] = 0$).

(b) 证明三维埃伦菲斯特定理

已知: $\frac{d}{dt} \langle \hat{Q} \rangle = \frac{1}{i\hbar} \langle [\hat{Q}, \hat{H}] \rangle + \langle \frac{\partial \hat{Q}}{\partial t} \rangle$

- 证明 $\frac{d}{dt} \langle \hat{r} \rangle = \frac{1}{m} \langle \hat{p} \rangle$, 取 $\hat{Q} = \hat{r}$, 有 $\frac{\partial \hat{r}}{\partial t} = 0$, $\hat{H} = \frac{\hat{p}^2}{2m} + V(r)$

$$\text{计算 } [\hat{r}, \hat{H}] = [\hat{r}, \frac{\hat{p}^2}{2m} + V(r)] = [\hat{r}, \frac{\hat{p}^2}{2m}] = \frac{1}{2m} [\hat{r}, \hat{p}^2]$$

其中 x 分量 $[\hat{x}, \hat{p}_x^2] = 2i\hbar \hat{p}_x$, 故 $[\hat{r}, \hat{p}^2] = 2i\hbar \hat{p}$

$$\text{故 } \frac{d}{dt} \langle \hat{r} \rangle = \frac{1}{i\hbar} \langle [\hat{r}, \frac{\hat{p}^2}{2m} + V(r)] \rangle = \frac{1}{i\hbar} \cdot \frac{1}{2m} \langle 2i\hbar \hat{p} \rangle = \frac{1}{m} \langle \hat{p} \rangle$$

- 证明 $\frac{d}{dt} \langle \hat{p} \rangle = -\langle \nabla V \rangle$, 取 $\hat{Q} = \hat{p}$, 有 $\frac{\partial \hat{p}}{\partial t} = 0$

$$\text{计算 } [\hat{p}, \hat{H}] = [\hat{p}, \frac{\hat{p}^2}{2m} + V(r)] = 0 + [\hat{p}, V(r)]$$

其中 x 分量 $[\hat{p}_x, V(x)] = -i\hbar \frac{\partial V(x)}{\partial x}$, 有 $[\hat{p}_i, f(r)] = -i\hbar \frac{\partial f}{\partial r_i}$

$$\text{故 } [\hat{p}, V(r)] = -i\hbar \nabla V$$

$$\text{因此, } \frac{d}{dt} \langle \hat{p} \rangle = \frac{1}{i\hbar} \langle [\hat{p}, \hat{H}] \rangle = \frac{1}{i\hbar} \langle -i\hbar \nabla V \rangle = -\langle \nabla V \rangle \text{ 得证}$$

(c) 三维海森堡不确定性原理

- 因 $\sigma_A \sigma_B \geq \frac{1}{2} |\langle [\hat{A}, \hat{B}] \rangle|$, $[\hat{r}_i, \hat{p}_i] = i\hbar$

$$\text{故 } \sigma_{r_i} \sigma_{p_i} \geq \frac{1}{2} |\langle i\hbar \rangle| = \frac{\hbar}{2}$$

$$\text{即 } \sigma_x \sigma_{p_x} \geq \frac{\hbar}{2}, \sigma_y \sigma_{p_y} \geq \frac{\hbar}{2}, \sigma_z \sigma_{p_z} \geq \frac{\hbar}{2}$$

- 位置或动量不同分量对易关系为 0, 如 $[\hat{x}, \hat{p}_y] = 0$, 因此 $\sigma_x \sigma_{p_y}$ 等无限制, 理论上可同时精确测量。